

ASWIN P T

AI/ML Engineer | Backend Developer

Phone: +91 88913 81713, +91 96455 99051 | **Personal Website:** codedbyaswinpt.vercel.app

LinkedIn: linkedin.com/in/aswinpt2004 | **GitHub:** github.com/Aswinpt2004 | **Email:** aswinsureshpt@gmail.com

PROFESSIONAL SUMMARY:

Machine Intelligence graduate student at Digital University Kerala with hands-on experience in federated learning, graph neural networks, differential privacy, and end-to-end AI engineering. Completed a 12-week industry internship delivering privacy-preserving AI systems for healthcare and financial fraud detection. Skilled in Python, PyTorch, FastAPI, and GPU accelerated model development. Driven by research rigour, production-grade engineering discipline, and a strong commitment to building AI systems that are both performant and privacy-aware

EDUCATION:

M.Sc. in Computer Science (Machine Intelligence)

Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram (2024 – 2026)

B.Sc. in Computer Science

College of Applied Science IHRD, Kozhikode (2021– 2024)

TECHNICAL SKILLS:

- **Programming:** Python, C, C++
- **AI/ML:** Deep Learning, Computer Vision, Data Analytics, Cognitive Computing, NLP, Federated Learning
- **Frameworks:** PyTorch, TensorFlow, OpenCV
- **Tools:** Google Colab, Kaggle, VS Code, PyCharm, Jupyter Notebook, Git, Google Cloud Platform
- **Databases:** PostgreSQL, SQL
- **Backend & Web :** Flask, Django, REST API Development, HTML, CSS, JavaScript

EXPERIENCE :

AI/ML Intern : Synbrains Solutions Pvt Ltd (Mar 2026 – May 2026)

12-week progressive internship covering production AI engineering, backend development, federated learning, and privacy-preserving ML across healthcare and financial domains.

PI-RADS Prostate AI Reporting System

- Contributed to a clinical AI reporting platform updated report layouts, implemented dropdown improvements, completed scoring, image handling, and sector mapping modules.
- Applied FastAPI backend integration and Git branching workflows for collaborative feature development.

Shear Wave Elastography Report Parser

- Developed an automated medical report extraction pipeline from specification documents with zero prior codebase.
- Built structured extraction pipelines using Pydantic data contracts and heuristic regex-based parsing to normalise raw PDF/DOCX report content into validated ReportDraft schemas.

PROJECTS :

FedGraph-Shield : Hierarchical Federated Graph Learning for Financial Fraud Detection Ongoing

- Designed a hierarchical federated graph learning framework allowing distributed banking institutions to collaboratively train a fraud detection model without sharing raw transaction data.
- Modelled financial transactions as directed heterogeneous graphs customers, merchants, and devices as nodes; payments as edges enabling detection of ring structures and circular payment chains invisible to tabular approaches.
- Coordinated federated aggregation via the Flower framework using a two-tier hierarchical FedAvg strategy with differential privacy composition for formal privacy accounting across communication rounds.

Dynamic Connectome Based Prediction of Reality Monitoring Failure Under Sleep Deprivation (Graph Attention Networks)

Python · PyTorch · fMRI Preprocessing · Graph Attention Networks

- Designing a dynamic connectome framework using resting-state fMRI to predict reality monitoring failure before overt behavioural symptoms appear with applications to psychosis-risk and fatigue research.
- Employing Graph Attention Networks (GATs) to identify informative prefrontal sensory connections associated with sleep-deprivation-induced network instability currently at the preprocessing and parcellation stage.

Adaptive Confidence Modeling for SHADER-Based 3D Human Body Shape Estimation

Python · PyTorch · OpenCV · SMPL

- Introduced a dynamic, image-driven confidence model replacing the original global- σ formulation in the SHADER monocular 3D shape estimation pipeline for loose-clothing scenarios.
- Achieved 20–25% lower confidence-prediction error on 9,656 dressed-human images, producing more stable SMPL estimation without additional data or architectural changes.

PETZZO : Comprehensive Pet Care Platform

Django · HTML · CSS · JavaScript · MySQL

- Built a full-stack platform connecting pet owners with verified hospitals, veterinarians, and pet shops, featuring AI-driven disease prediction, real-time pet tracking, and appointment booking.
- Implemented pet health record and vaccination management, pet adoption workflows, and streamlined pet shop operations within a single integrated system.

Manglish WhatsApp Chatbot (Llama-3 / Ollama)

Node.js · Llama-3 · Ollama · Hybrid AI

- Built an offline Malayalam English chatbot combining rule-based intent analysis, similarity-based response retrieval, and local Llama-3 inference enabling cloud-independent real-time regional language messaging.

AI-Powered E-Commerce Storefront & Catalog Generation Agent

Python · Agentic AI · LLMs · Vision-Language Models · FastAPI

- Developed an Agentic AI platform that converts natural language prompts into fully generated e-commerce storefronts by autonomously creating store branding, category hierarchy, landing pages, and website content.
- Built a multimodal catalog generation pipeline leveraging Vision-Language Models (VLMs) and LLMs to extract product information from images and generate structured catalogs with AI-generated titles, descriptions, specifications, categories, and SEO-optimized metadata.

KuTu: AI-Powered Tour Guide Marketplace

Next.js · React · TypeScript · Tailwind CSS · AI Agents · Gemini API

- Built an AI-powered travel marketplace that connects travelers with verified local tour guides through fixed packages and a reverse bidding system, enabling users to create personalized trip requests based on destination, budget, and preferences.
- Developed an intelligent trip planning workflow using AI-generated itineraries, questionnaire-based travel customization, guide matching, and real-time offer negotiation, delivering a personalized booking experience similar to Uber for local tourism.